

1 (a) Define *impulse*.

Product of (average) force and the time that the (average) force acts on the body [1] (or time of impact)

[1]

- (b) In a car test, a car with a dummy driver and passenger, moving at a speed of 6.9 m s^{-1} , collides head-on into a wall. The mass of the car is 1250 kg, the mass of the driver is 85 kg and the mass of the front passenger is 65 kg. The average deceleration of the car as it comes to a stop is 48 m s^{-2} . Both passenger and driver have their seat belts tightly fastened.

For the impact,

- (i) determine the magnitude of the average force exerted on the car and its occupants.

$$\text{average force} = ma = (1250+85+65) (48) = 67200 \text{ N}$$

average force = N [1]

- (ii) determine the magnitude of the impulse caused by the force.

$$\begin{aligned} \text{Impulse} &= \text{change in momentum} \\ &= mv - mu \\ &= 0 - (1250+85+65) (6.9) \quad [1] \\ &= -9660 \text{ Ns} \\ \text{Ans: } 9660 \quad [1] \end{aligned}$$

impulse = N s [2]

- (iii) Hence, calculate the time taken for the car to come to a stop.

$$\begin{aligned} \text{By Impulse - momentum theorem,} \\ Ft &= mv - mu \\ t &= 9660/67200 \quad [1] \\ &= 0.144 \text{ s} \quad [1] \end{aligned}$$

time = s [2]

- (iv) Assuming that the average deceleration remains the same, state and explain how your answer in (b)iii will change (if any) when the total mass of car and occupants has doubled.

No change [A1].
Since the change in velocity and the deceleration remains unchanged ($a = \Delta v / \Delta t$) [M1]
Or since the impulse (the change in momentum) and the average force both doubled, the time taken remains unchanged [M1]

[2]

2. A fixed mass of an ideal monatomic gas undergoes a cycle ABCA of changes as shown in